

GlobalFinTech Inc. | Linux Operating System Evaluation and Recommendation

Document Overview

This document provides a high-level evaluation of the security, stability, and performance of the Linux operating system (OS) to inform its feasibility for use in GlobalFinTech Inc.'s distributed cloud-based infrastructure. The company's main requirements and operational constraints are detailed, followed by the overall recommendation of the report. In the Evaluation section, details are given for security mechanisms, stability features, the Linux architecture, handling of system resources, failure metrics, support for distributed environments, and key performance metrics.

This evaluation was conducted through a review of academic literature, industry reports, and technical documentation addressing the security architecture, reliability, and performance characteristics of the Linux OS. Additionally, companies that currently utilize Linux in their systems were identified, along with up-to-date best practices and recommendations for hardening the Linux OS. The final recommendation was made based on a holistic view of the benefits and drawbacks of Linux specifically for GlobalFinTech Inc.'s use case.

Company Context

GlobalFinTech Inc. is a growing cloud-based digital payment processing platform for small and medium-sized businesses to accept online and mobile payments.

Primary Services

- Online and mobile payment processing
- International money transfer and currency exchange services
- Fraud detection and transaction monitoring
- Hosted financial data application programmer interface (API) for secure connection to partner services (e.g. banking apps, mobile wallets)

Scale of Operations

- Serves approximately 1 million active users through 100 thousand business clients
- Processes 3-5 million transactions per day, with peak loads during business hours in North America
- Stores and processes large volumes of financial and personal data

Compliance Requirements

- Gramm-Leach-Bliley Act - protection of consumer financial data
- Payment Card Industry Data Security Standard - storing and processing of card information
- Bank Secrecy Act - detection and prevention of fraud and money laundering

Operational Constraints

- Requires 99.9% uptime to ensure availability
- Maintenance of encryption and monitoring to meet compliance and prevent data breaches
- High-performance resource management to handle concurrent transactions with no latency
- Must support rapid scaling during peak transaction spikes

Recommendation

The final recommendation of this report is: *Linux is an appropriate operating system choice for GlobalFinTech Inc.'s distributed cloud-based infrastructure.*

The modular architecture of the Linux kernel, extensive open-source ecosystem, and widespread industry adoption make Linux a strong candidate for supporting GlobalFinTech Inc.'s scalable and security-focused infrastructure requirements. While Linux provides confidentiality, integrity, and availability using core mechanisms, the key benefit of Linux is the additional security that can be gained through configuration and external software tools. For a distributed cloud-based system, Enterprise RedHat with SELinux for enhanced security controls and Kubernetes to manage containerization leads to a highly scalable, fault-tolerant, secure infrastructure that is capable of growing with GlobalFinTech Inc.'s customer base.

However, Linux systems require experienced administrators to properly configure and maintain security policies, which may increase initial implementation complexity compared to more tightly managed operating systems such as Windows. Additionally, the growing popularity of the Linux OS has led to an increase in efforts by malicious actors to discover vulnerabilities, potentially raising the risk of the system.

Despite these drawbacks, Linux remains a strong and sensible OS choice for GlobalFinTech Inc. due to its modular architecture, security software ecosystem, and support for distributed and containerized environments.

Evaluation

Security Principles and Mechanisms

The Linux OS and kernel provide numerous security mechanisms. Discretionary access controls (DAC) and mandatory access controls (MAC) ensure proper authentication and authorization is met for data access (Silberschatz et al., 2023). Networking security is configured through iptables or the uncomplicated firewall (UFW), which restricts the flow of inbound and outbound traffic according to defined policies (Waleed, 2023). Virtualization and restriction features such as control groups (cgroups) and namespaces allow for fine-grained tuning of resource and process allocation to maintain system integrity (Zehra, 2024). The secure computing mode—or seccomp—feature of the Linux kernel restricts the types of system calls a process may make, enhancing security by limiting access to the kernel (Zehra et al., 2024).

In addition to the security mechanisms shipped by default, Linux has a rich pool of security applications in the open-source community as well as at the enterprise level. SELinux or AppArmor provide granular MAC with security policies (Albatli et al., 2023; Day 2025). OpenSSL, Snort, pluggable authentication modules (PAMs), extended Berkeley Packet Filter (eBPF), and network mapper (nmap) are other Linux security tools for hardening the OS (Zehra et al., 2024).

However, due to its modular and highly configurable architecture, the security posture of a Linux system depends heavily on proper system configuration and administrative practices. As shipped with default settings, the Ubuntu distribution was shown to pass only 42.44% of measured controls in the Center for Internet Security (CIS) audit (Sedano et al., 2021). Proper hardening, for example of the `sysctl.conf`,

passwd, shadow, and fstab files, as well as iptables or UFW rules, is vital for securing the system (Day, 2025).

Stability and Reliability Features

Linux is known to be a stable and reliable OS due to its robust architecture and open-source community that is able to quickly implement patches. Additionally, Linux's support for distributed and scalable systems and high availability/failover tools such as Pacemaker and Corosync ensure system availability and stable uptimes through redundancy mechanisms (Zehra et al., 2024). Linux filesystems such as ext4 and XFS include journaling, which records filesystem changes before they are committed, allowing the system to recover quickly from crashes or power failures. Many enterprise distributions provide the ability to apply live kernel patches and security updates without the need for a reboot (Day, 2025).

However, Linux has faced many known issues in recent years. For example, a kernel driver vulnerability allowed for a remote attacker to execute arbitrary code using crafted packets in CVE-2015-8812 (Day, 2025). Another critical vulnerability was CVE-2017-18017, where overflow issues in accepted transmission control protocol (TCP) packet headers left systems vulnerable to denial of service (DoS) attacks (Day, 2025).

Despite these examples of known failure points, the open-source nature of Linux allows for the rapid development and application of patches without the need to rely on an external party to roll out fixes (Wang et al., 2024). This makes Linux an ideal choice for maintaining a stable and reliable distributed cloud infrastructure. No system will ever be perfectly secure, so a system is better judged by how it responds to and recovers from failures rather than how many occur.

Integrity, Confidentiality, and Availability

The safeguarding of information from unauthorized access and maintenance of confidentiality and integrity in the Linux OS is mainly ensured by DAC and MAC. MAC providers are often shipped with enterprise Linux OSes, such as SELinux for RedHat and AppArmor for Ubuntu distributions (Day, 2025). PAMs also support enhanced authentication methods. The integrity of information also relies on access control, with the addition of checksums and version control to identify when data has changed (Adekotujo et al, 2020). Logging utilities, such as auditd and syslog, provide additional assurance of data integrity.

To ensure confidentiality, data should be encrypted both at rest and in transit using appropriate methods. OpenSSL and VeraCrypt are commonly used for this purpose. Availability in Linux systems is supported through redundant services, automated failover mechanisms, and distributed service architectures, ensuring that system operations can continue even if a node becomes unavailable. Once again, the assurance of the confidentiality, integrity, and availability (CIA) triad ultimately falls on the configuration of the numerous Linux tools and utilities at the administrator's disposal. Linux provides a lot of freedom in how a system's security needs will be met, but also leaves a lot of room for potential vulnerabilities due to negligence or misconfiguration.

OS Architecture

The Linux OS consists of 4 main layers: applications and utilities, the shell, the kernel, and the underlying hardware. The kernel itself is composed of the process scheduler, memory management unit, the virtual

file system (VFS), networking subsystem, and interprocess communication (IPC) unit (GeeksForGeeks 2023). The modular nature of the Linux kernel allows individual subsystems or drivers to be loaded or updated independently, improving maintainability and allowing security updates without requiring major system redesign.

At the hardware level, support for trusted platform modules (TPMs) is pivotal (Zehra et al., 2024). TPMs and hardware security modules (HSMs) “securely store encryption keys and credentials, ensuring their protection even in the event of system compromise” (Zehra et al., 2024).

The inherent decentralized and modular approach of the Linux kernel places an emphasis on flexibility and configurability (Adekotujo et al., 2020). Options such as Unified Extensible Firmware Interface (UEFI) SecureBoot—which requires cryptographically signed firmware and kernels—and kernel lockdown mode implement security measures at incredibly low levels within the system for protecting system integrity (Day, 2025). Additionally, the ability to implement live kernel patches is critical for system availability and stability (Day, 2025).

The shell exposes critical utilities for system administrators to harden the OS. Firewall rules, access control, networking configurations, process management, scanners, and logging are all available through the shell (Singh, 2024).

Ensurance of system efficiency and reliability at the application layer is a result of its separation from the kernel layer. With the kernel’s orchestration, user-level applications are served with the appropriate execution time, memory, and resources to accomplish their tasks. The implementation of namespaces, cgroups, and the VFS ensure that processes do not have more access than necessary (Zehra, 2024).

System Resources

Recent Linux kernels utilize the Earliest Eligible Virtual Deadline First (EEVDF) scheduler while prior versions utilized the Completely Fair Scheduler (CFS) (*EEVDF Scheduler — the Linux Kernel Documentation*, 2024; Silberschatz et al., 2023). In either case, the scheduler “aims to distribute CPU time equally among all runnable tasks with the same priority,” supporting both long-running background tasks and latency-sensitive real-time tasks (*EEVDF Scheduler — the Linux Kernel Documentation*, 2024). Support for both multithreading and hardware threading (hyperthreading) further enhances system performance through the kernel’s one-to-one threading model (Silberschatz et al., 2023).

Memory management is handled through virtual memory and paging, maximizing efficient memory allocation (Silberschatz et al., 2023). Additionally, the Linux kernel is Non-Uniform Memory Access (NUMA) aware, allowing for physically close memory nodes to be used optimally (Silberschatz et al., 2023). Linux also utilizes cgroups to limit and prioritize central processing unit (CPU), memory, and input/output (I/O) usage across different processes or containers. This ensures that critical services maintain adequate resources even during peak workloads (Zehra et al., 2024).

External resources, such as devices and disk, are also managed using schedulers. It should be noted that the Linux kernel *does not* detect or prevent deadlocks in user or kernel space (Silberschatz et al., 2023). The rationale behind this is that implementing deadlock identification, such as a resource-allocation graph (RAG), would degrade system efficiency or performance more than would be gained by preventing deadlocks, since they occur infrequently.

Failure Metrics

Intrusion prevention and intrusion detection (IPID) systems such as Snort provide real-time monitoring capabilities, allowing for dynamic process management to recover from attacks that may compromise the system (Singh, 2024). Linux supports thorough auditing and logging to track failure metrics such as mean time between failures (MTBF). System performance can be monitored live using tools such as top, htop, turbostat, and pqos. Additional reliability metrics commonly tracked in Linux systems include mean time to recovery (MTTR) and service availability percentage, which measure how quickly systems recover from failures and how consistently services remain accessible. These metrics are particularly important in financial transaction systems where downtime directly impacts customer trust and regulatory compliance.

Linux servers have historically demonstrated high uptime and operational resilience in enterprise and cloud environments. The main mechanism for providing fault tolerance is through a distributed environment that allows for redundancy in services that can effectively load balance if a single node is compromised (Singh, 2024). Many enterprise environments also implement automated monitoring platforms such as Prometheus or Nagios, which collect performance metrics and trigger alerts when system thresholds are exceeded (Waleed, 2023).

Distributed Environments

For a cloud-based and scalable infrastructure, virtualization and containerization is a must. Linux provides excellent support for containerization orchestration platforms such as Kubernetes (Waleed, 2023; Zehra et al., 2024). The Linux kernel supports kernel-based virtual machines (KVMs) which provide efficient communication, resource sharing, and processing synchronization. Clustering support handles service discovery, load balancing, automatic scaling, configuration management, and health monitoring to effectively use the underlying system resources (Singh, 2024).

The main benefits of effective containerization and virtualization are increased resource efficiency, flexibility, isolation, cost savings, and improved performance (Singh, 2024). Linux containerization technologies such as Docker rely heavily on kernel features like namespaces and cgroups to isolate applications while sharing the same kernel, improving efficiency compared to traditional virtual machines (Singh, 2024). In distributed systems, Linux nodes can also participate in load balancing and horizontal scaling architectures, enabling additional compute nodes to be added dynamically as transaction volumes increase.

Key Performance Metrics

Linux provides utilities for measuring key performance metrics such as top and htop. These tools provide realtime feedback on CPU utilization, memory usage, and disk performance as well as system uptime. In addition to overall system efficiency, top and htop provide metrics at the individual process level, allowing administrators to track down inefficient processes or potential malicious users and connections (Singh, 2024). Other commonly monitored performance indicators include network throughput, disk I/O latency, and process scheduling latency, all of which can impact the responsiveness of financial transaction systems. Daemons such as auditd and features like syslog enable advanced system monitoring with robust and traceable output (Day, 2025). To maintain the integrity of logs, files must be stored in a centralized location separate from the host system (Day, 2025). (Singh, 2024).

Linux is commonly used in distributed cloud environments and is trusted by major companies such as Amazon Web Services (AWS), Google Compute Engine (GCE), and Azure Virtual Machines (AVM) (Singh, 2024). These large-scale deployments demonstrate Linux's capability to maintain consistent performance under highly concurrent workloads, making it well suited for high-volume financial processing environments. However, this wide acceptance by industry has led to a recent switch from Windows to Linux for being the most targeted OS by malicious attackers (Singh, 2024).

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